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Explainable Artificial Intelligence (AI) for Medical Imaging: A Framework for Bridging the AI Trust Gap

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Abstract

Artificial intelligence (AI) is increasingly used in healthcare but often lacks clinician and patient trust. Explainable AI (XAI) aims to clarify predictions and to make AI decisions more transparent, interpretable, and clinically actionable. Yet, current methods fall short. In this Perspective, we argue that, for XAI to be clinically useful in medical imaging and to build trust with clinicians, it must satisfy three guiding principles: technical robustness, adaptation to end users, and alignment of explanations with the specific clinical task. We introduce a conceptual framework, incorporating these principles, to guide future XAI design and deployment based on expectations and shared responsibilities for developers, vendors, and healthcare institutions. By ensuring robustness, personalizing outputs, and aligning explanations with use cases, XAI can move beyond one-size-fits-all approaches to task- and user-centered design, to support effective and trustworthy AI adoption in healthcare.

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Highlights

- The rapid development and deployment of AI solutions have far outpaced methods to explain their outputs, generating a trust gap for clinicians and patients.
- To address this gap, future XAI method development must be technically robust and adapt to individual users' preferences and workflow needs.
- XAI methods will also need to provide explanations that match the purpose of the clinical task at hand.

Introduction

Recent technologic advances have rapidly shifted perceptions regarding artificial intelligence (AI) in medicine. Once viewed as a distant possibility, AI is now seen by many as an inevitable part of future clinical practice. This transformation is evident in regulatory trends: of all AI-enabled medical devices cleared by the U.S. FDA, 78% were cleared after 2019 [1]. The rapid development and deployment of AI solutions have far outpaced methods to explain their outputs, generating a gap in AI trust by clinicians and patients. Indeed, despite increasing interest in AI in medicine, widespread trust has remained elusive. For example, a 2022 Pew Research Center survey found that 60% of Americans would feel uncomfortable with healthcare providers relying on AI for their own care [2].

To address this gap, explainable AI (XAI) methods have been developed to shed light on how AI systems arrive at their predictions [3]. Yet, in radiology, which accounts for 77% of all current FDA-cleared AI-enabled medical devices, the promise of XAI to build trust with end-users has been met with skepticism. Critics highlight concerns about the clinical utility and trustworthiness of explanations of XAI in medical imaging [4]. These concerns are compounded by limited and conflicting evidence on how clinicians are impacted by and engage with XAI outputs in radiology practice [5,6]. Together, these limitations raise doubts about the feasibility of a one-

size-fits-all approach to XAI in medical imaging and highlight the need, instead, for domain-specific tailored approaches.

In this Perspective, we argue that, for XAI to be clinically useful in medical imaging and to build trust with clinicians, it must satisfy three guiding principles: technical robustness, adaptation to end users, and alignment of explanations with the specific clinical task. We introduce a conceptual framework, incorporating these principles, to guide future XAI design and deployment based on expectations and shared responsibilities for developers, vendors, and healthcare institutions.

Technical Robustness: XAI Should Identify Clinically Important Features, Describe How They Influence Predictions, and Pass Sanity Checks

For XAI to be clinically useful in medical imaging, it must be technically robust. We define technical robustness (**Fig. 1A**) as the ability of XAI methods to accurately highlight features that are meaningful to clinicians, precisely convey the sense in which these features are important for AI predictions, and respond appropriately to perturbations that should or should not affect their outputs (i.e., sanity checks) [7,8]. Despite widespread use of these criteria in the medical AI literature, many XAI methods fall short.

XAI methods, such as saliency maps, are commonly used in both academic research and commercial products [3] to indicate ROIs in medical images. However, evidence suggests that these methods often fail to reliably localize true abnormalities [4,8,9]. Arun et al. demonstrated this phenomenon through an evaluation of several common methods for saliency map generation in pneumothorax localization in which none of the methods outperformed dedicated pneumothorax segmentation models [8]. More concerningly, all but one of the tested methods (explanation with ranked area integrals) failed to outperform an intentionally low baseline of the simple mean of the reference-standard pneumothorax segmentations. In other words, the act of simply highlighting the same location on each chest radiograph was overall more accurate than

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the saliency methods in generating separate maps tailored to each unique radiograph. Additionally, even for correct AI predictions, saliency maps only highlight the parts of an image that influenced the model's prediction, which may or may not correspond to the precise clinical finding (e.g., a pneumonia opacity on a chest radiograph). This mismatch reflects a gap between the technical meaning of these visual explanations and their perceived meaning by clinicians. This disconnect is not just academic. Work in human-computer interaction has shown that misleading XAI outputs in radiology can degrade physician diagnostic accuracy, particularly in diagnostic imaging tasks [5]. This risk may be exacerbated by the phenomenon known as "simple trust," whereby clinicians place undue confidence in AI outputs, even when those outputs are flawed [5].

A second aspect that is paramount for XAI to be useful in clinical settings is precise communication of the meaning of the reported important features. Although many XAI approaches highlight input features as important, the meaning that such features convey can differ significantly. For example, many popular XAI methods use forms of gradient information to convey importance [10]; such information may or may not reflect actual relevant important features. In contrast, other methods denote features as important if they are sufficient and/or necessary for producing specific outputs [11]. An understanding of the differences between these complementary notions of feature importance is crucial, as they may convey dramatically different meanings about the task and predictive model under consideration. Sufficient features imply that their presence alone suffices for making the given prediction (i.e., regardless of other features that may be present); necessary features imply that, without their presence, the predictive result would be different [11]. Consider an AI model that predicts the presence of right lower lobe pneumonia on chest radiograph and outputs a heatmap highlighting a right lower lobe pneumonia that received model attention. This feature would be sufficient if the model's original pneumonia prediction would be preserved even if the influence of the rest of the image' on the prediction is removed. In comparison, the same feature would be necessary if its removal or perturbation would result in the model returning to an uncertain prediction [12]. Other methods employ game-

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theoretic notions to attribute importance to different features through hypothesis testing [13]. Unfortunately, a nonnegligible gap exists between the technical implications of these notions and the way in which these tools are applied in real-world clinical settings.

A third dimension of technical robustness involves whether XAI outputs change or remain stable in logical and expected ways when the underlying model is altered (i.e., sanity checks). Developers can employ functionally grounded metrics to quantify this dimension before human evaluation, [14]. For example, feature importance stability coefficients measure consistency across similar inputs, while perturbation sensitivity indices help to ensure that the explanation degrades appropriately when the input is corrupted. Many XAI techniques used in medical AI fail these basic checks. For example, saliency maps have been shown to remain visually similar between neural networks trained to diagnose a disease on medical images and randomly initialized neural networks [4,8] that ostensibly lack any medically relevant knowledge. These limitations have real-world implications. In one study [15], radiologists were randomized to diagnose intracranial aneurysms on CT examinations using either a high-performance AI model or a placebo sham AI model with near-zero diagnostic sensitivity. Both systems produced saliency maps. Strikingly, radiologist performance was no worse when using the sham AI system, suggesting that both models' saliency maps were functionally similar despite vastly different underlying model performance and initialization.

Together, these findings underscore the importance of technical robustness in XAI development for medical imaging. Without accurate localization and demonstrable robustness to technical sanity checks, XAI methods risk producing explanations that are, at best, clinically uninformative and, at worst, actively misleading.

Adapt: XAI Needs to Be Specific to End Users

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In medical imaging, XAI is ultimately built to serve human decision-makers: clinicians who vary widely in their expertise, cognitive styles, and workflow demands. Accordingly, XAI must be specific to these clinical end users and their unique preferences (**Fig. 1B**). Yet, most current XAI systems are static, offering a single uniform explanation to all users regardless of individual needs or preferences. This mismatch can reduce the effectiveness of AI assistance and potentially lead to adverse clinical outcomes.

End users of AI tools for medical imaging span a spectrum of training backgrounds from subspecialist radiologists to general practitioners, physician assistants, and nurses. These differences shape how users interpret AI outputs and the kinds of explanation that they find most useful. Even among radiologists, studies have shown meaningful variation in individual responses to AI assistance [5,6], suggesting lack of a universal optimal format for explanation. Some may benefit most from visual cues like saliency maps, whereas others may prefer structured textual explanations that mirror conventional radiology reports. These preferences may reflect cognitive styles; cognitive biases due to personality, job experience, or life background; or contextual needs including time constraints during high-volume interpretation sessions.

Emerging evidence highlights the risks of poor alignment between XAI output and user needs. Explanations that are too simplistic, overly detailed, poorly timed, or misaligned with clinical expectations can lead to confusion or diagnostic errors [16]. In high-stakes environments like medical imaging, even small mismatches can have significant consequences.

Fortunately, XAI does not need to remain static. Just as AI models can learn from data to optimize clinical performance, they can also be trained to adapt their explanatory behavior to individual users. Techniques such as reinforcement learning from human feedback (RLHF) or direct preference optimization (DPO) offer scalable pathways to personalization [17], as demonstrated by their central role in adapting modern commercial large language model outputs (e.g., ChatGPT [OpenAI], Gemini [Google]) towards human-preferred responses [18,19]. While still in the early stages of development for radiology, these techniques have been used to improve

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image segmentation and the factual accuracy of radiology report generation [20,21]. By dynamically tailoring explanations, whether in modality, complexity, or timing, AI systems can better integrate into clinical workflows and support informed decision-making. Nonetheless, not all clinical tasks will require user customization. For straightforward highly constrained tasks such as detecting pneumoperitoneum or measuring an aneurysm, a simple static explanation (e.g., a single bounding box or anatomic segmentation) is often sufficient. Institutions should evaluate the need for customization based on the clinical complexity of the task, the diversity of the intended user base (e.g., single vs multiple radiology subspecialties or single vs multiple medical specialties), and likelihood that a one-size-fits-all output would disrupt workflow.

The shift from static to adaptive XAI also introduces significant regulatory and practical challenges. To date, the vast majority of FDA-cleared AI-enabled medical devices have followed a linear deployment model whereby model parameters are locked or frozen at the time of clearance to ensure predictable performance. Transition to dynamic deployments whereby explanations may evolve over time requires robust infrastructure for continuous monitoring and a shift in regulatory oversight toward frameworks like the predetermined change control plan (PCCP), which allows pre-specified automated updates while maintaining safety [22,23]. Importantly, developers, vendors, and healthcare institutions share the responsibility for adaptation. Developers should provide the technical architecture to generate appropriately structured feedback data for XAI output preference tuning (e.g., generating at least two XAI outputs for a single chest radiograph and asking radiologists to choose their preferred output), while vendors should provide the necessary technical support to organize the data and facilitate preference training runs. Concurrently, healthcare institutions must manage the local implementation, provide feedback, and use their clinical expertise to monitor XAI outputs, actively evaluating the reliability of XAI adaptations by ensuring that they continue to pass technical sanity checks and do not negatively impact clinician diagnostic accuracy during local validation.

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To fulfill its promise in medical imaging, XAI must move beyond generic outputs. Instead, it should aim to meet individual clinicians in delivering the right explanation in the right form and at the right time. However, personalization alone is insufficient for ensuring interpretability and safe use; as described further in the next section, the XAI explanation format must also match the clinical task and model output.

Match: XAI Explanations Should Be Specific to the Clinical Task

Beyond providing explanations tailored to individual end users, XAI in medical imaging must be tailored to the underlying clinical task and the form of model output. Accordingly, a variety of discrete XAI methods are employed in practice including in FDA-cleared AI products. These methods can be categorized based on their user interface [3] and include localization-based methods [3,5,8–10] (e.g., highlighting ROIs on a saliency map), quantitative-based methods [3] (e.g., reporting an ascending thoracic aortic aneurysm as being at risk for rupture based on a diameter >5 cm), semantic-based methods [3] (e.g., classifying a lung nodule as malignant due to its spiculated shape), and exemplar-based methods [3,5] (e.g., providing the user with similar cases for comparison). In some cases, multiple XAI methods are combined. For instance, an AI tool for assessing lung nodule malignancy risk may use both semantic and quantitative explanations. These methods aim to collectively explain three central questions (what, why, and where) of an AI model's output (**Fig. 2**).

Because end users differ in expertise and workflow context, task-appropriate explanation choices also affect how reliably users can interpret with and act on AI outputs. However, the optimal choice of XAI method depends on the type of output produced by the AI model (**Fig. 1C**). For example, models with discrete outputs (e.g., a disease classifier with such categories as "pneumonia," "pneumothorax," or "normal") may benefit from methods that highlight features leading to a specific classification. In this case, a saliency may show a region of right middle lobe consolidation to support a pneumonia diagnosis. In contrast, regression models, such as those

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predicting a continuous variable like pediatric bone age, require a different approach. For these models, a semantics-based XAI method may be more appropriate. These methods indicate the manner in which specific imaging features influenced the prediction (e.g., whether certain growth plates are open or closed) and show how changes in those features affect the output on a spectrum (recognizing that the task of bone age prediction lacks a discrete abnormality to localize).

An understanding of the preferred XAI method to use for a given clinical task is critical to facilitate clinically safe and trustworthy deployment of AI in medical imaging. Much of the current literature on XAI in medical imaging focuses on classification tasks, which commonly use local pixel-level localization methods like saliency maps or bounding boxes [4–10]. However, these methods may not be suitable for other diagnostic tasks such as bone age prediction, which lack a discrete abnormality to highlight. Moreover, given the diversity of XAI methods used in FDA-cleared commercial tools, it is essential that clinical leaders responsible for AI tool procurement understand these nuances. Although the developer is responsible for providing a suite of task-appropriate XAI outputs, the procuring institution is primarily responsible for selecting the specific configuration that best integrates with their local clinical workflows. Because successful task-matching is highly context-dependent, measurement of success requires application-grounded evaluation [14]. In radiology, this process requires that the XAI be rigorously evaluated by domain experts (e.g., subspecialty radiologists) executing precise clinical tasks rather than relying on generalized user testing. Such an approach ensures that the chosen matching method enhances, rather than disrupts, the diagnostic process for the specific use case. These task-specific considerations complement user-centered personalization and underscore why XAI design choices should be evaluated both technically and clinically before deployment.

The most appropriate XAI approach for a given task is the one that matches the AI model's primary objective to its intended downstream clinical action. We recommend a three-part heuristic for identifying the optimal approach. The first objective is localization: for tasks requiring visual

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search, targeted biopsy, or surgical planning, models should be matched with discrete localization-based XAI (e.g., bounding boxes or precise segmentations). The second objective is quantification: when an AI model's output determines threshold-based triage or staging based on established guidelines (e.g., measuring ventricular volume or aortic dilation), it should be paired with quantitative or semantic XAI that explicitly contextualizes the measurement against clinical thresholds. The third objective is holistic classification and regression: for tasks for which a single distinct abnormality is ill-defined or represents a weighted classification of multiple findings (e.g., bone age estimation), localization of an abnormality is insufficient. Models fitting the third objective benefit most from exemplar-based or counterfactual XAI; such methods help clinicians evaluate the output by comparing the current imaging to known reference cases or by illustrating how specific semantic features influence the final AI prediction.

Importantly, for more complex diagnostic formulations like lung cancer staging, multiple XAI types may be required to adequately explain the model's output. For example, a comprehensive AI tool for TNM staging would first need to employ localization-based XAI (e.g., precise 3D segmentations) to identify the primary pulmonary lesion, suspicious mediastinal lymph nodes, and potential osseous metastases. Next, the tool would require quantitative XAI to communicate these findings' measurements, such as the primary tumor's maximum diameter or a lymph node's short-axis diameter, and directly contextualize such values against established clinical size thresholds for T staging. Finally, to justify the overarching prediction of the final clinical stage, the system may use semantic XAI to highlight qualitative features like pleural invasion or to describe the relation of lymph nodes and metastasis to the primary lesion (e.g., contralateral mediastinum) for purposes of N and M staging, respectively. By conducting this task analysis before development or procurement, institutions and vendors can move beyond trial-and-error and systematically select XAI tools that align with medical workflows.

Interdependencies in Implementation

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While it is useful to view the three presented XAI principles as conceptually separate, they are not independent in real-world practice and should not be pursued in parallel without constraint. We propose a hierarchic and iterative relationship among them. Technical robustness is a prerequisite: explanations must first be faithful to the model and pass sanity checks before they are adapted or matched. Task matching then constrains the space of acceptable explanations by aligning outputs with the clinical objective and downstream decision-making requirements. Adaptation to end users operates within these constraints, tailoring the presentation and format of explanations without altering their underlying technical validity or task alignment. In this sense, adaptation is bounded by both technical robustness and task matching and should not be used to compensate for deficiencies in either. This prioritization is designed to help prevent scenarios in which persuasive but technically invalid or task-inappropriate explanations degrade clinical performance.

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Recommendations

To successfully translate XAI from research into clinical practice in medical imaging, XAI methods must go beyond theoretic appeal. They must be technically robust, responsive to the needs and contexts of individual end users, and aligned with specific clinical tasks and model outputs. Given variability in clinical workflows, imaging modalities, and user expertise, XAI systems should not only offer explanations but also enhance clinicians' ability to critically evaluate and act on them.

Table 1 summarizes our recommendations to guide the development and deployment of XAI in medical imaging, as described in the remainder of this section.

Ensure Technical Robustness

AI developers and regulatory agencies should adopt standardized protocols for assessing the technical robustness of XAI methods, requiring XAI to correctly identify features of the clinical

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abnormality and convey their meaning; to identify features that sufficiently explain the model's predictions and remain stable regardless of the presence or absence of other image features; and to identify features that are correlated with the model's prediction. Collectively, these criteria can help ensure that XAI methods are robust against input perturbations and model variations, thereby maintaining fidelity and reliability across diverse clinical scenarios. We recommend that developers use functionally grounded evaluation paradigms [14] to translate these criteria into measurable standards by employing computational proxy metrics and sanity checks without requiring humans to validate XAI fidelity before clinical testing. To achieve this aim, developers should move beyond traditional post-hoc saliency maps. One potential strategy is to develop inherently interpretable AI architectures such as concept bottleneck models that are trained to predict intermediate clinically recognizable concepts (e.g., pleural effusion or cardiomegaly) before outputting a final diagnosis [24]. Another approach is to train models to generate an XAI output for any positive prediction of a finding as part of its core objective function (e.g., a fracture detection model generating a segmentation map of fracture lucency alongside any positive prediction). Additionally, integration of anatomic constraints directly into loss functions during model training can help ensure that generated explanations align with medical reality and avoid highlighting irrelevant artifacts [25,26].

Design for Adaptive Personalization

AI systems should include adaptive interfaces that tailor explanations to individual users considering clinical role, cognitive style, and workflow context. Although still in the nascent stages of application in radiology, machine-learning techniques such as RLHF or DPO may provide a foundation for this personalization [17]. Developers must work closely with regulatory bodies to define clear boundaries for these adaptive systems including through the creation of PCCPs in accordance with FDA guidance from 2026 [23]. Under such plans, developers describe the intended modifications; the associated method for developing, validating, and implementing those

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modifications; and an assessment of their impact. Through this process, developers reduce the risk that personalized tailoring will compromise the AI device's underlying technical robustness or clinical safety.

Match XAI Explanations to Task and Output

XAI methods should be explicitly selected and designed based on the nature of the clinical task (e.g., classification vs regression) and type of model output. Alignment of explanation format with clinical expectations enhances interpretability and utility.

Final Considerations

In real-world practice, implementation of the currently presented principles may introduce tensions. When conflicts arise, we recommend giving priority to XAI that is technically robust and appropriately matched to its clinical task, followed by user adaptation operating within these constraints.

Collaboration among developers, vendors, and healthcare institutions will be essential for successful XAI deployment. We recommend that developers focus on technical robustness by ensuring high-fidelity sanity-checked outputs. In contrast, the match and adapt phases represent a partnership: developers provide a selection of task-appropriate XAI outputs and technical architecture for collecting user feedback data for subsequent preference tuning; vendors must provide the technical support in understanding and selecting XAI tools and facilitating preference training; and healthcare institutions must select the most task-appropriate XAI output type and determine its clinical utility while providing consistent feedback and leading local validation and monitoring of XAI outputs. This shared responsibility ensures that XAI is not just a technical feature but a clinically integrated tool that meets the safety and utility standards of a specific clinical environment.

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XAI holds tremendous potential to close the trust gap between clinicians and AI by making AI decisions more transparent, interpretable, and clinically actionable. Realizing this promise will require a shift from generic one-size-fits-all explanations to task- and user-centered design. Additionally, while clinician-facing XAI justly remains the primary focus of development given clinicians' ability to best discern XAI outputs in a given context, little research explores the impact of XAI on patient trust (e.g., the incorporation of supplemental XAI explanations into physician-patient discussions of a new diagnosis). Whether direct interaction with patients will yield additional benefits in decreasing the AI trust gap among patients remains an open question. As the medical AI field matures, a thoughtful and evidence-driven approach to XAI will be essential to improve the user experience and ensure safe and effective AI integration into healthcare.

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TABLES

Table 1. Recommendations to guide the development and deployment of explainable AI (XAI) in medical imaging.

Recommendation	Rationale	Examples
Ensure technical robustness of XAI methods.	Previous XAI methods have been shown to lack trustworthiness and robustness. Matching theoretic clinical usefulness with strong theoretic foundations is critical to build useful XAI methods.	- Ensure that methods accurately represent internal states of model predictions. - Move toward inherently interpretable AI architectures (e.g., concept bottleneck models) or models that are trained specifically to provide an output along with their primary prediction (e.g., generation of a bounding box around a consolidation to accompany a positive pneumonia prediction) rather than relying solely on post-hoc explainability [24]. - Develop models that resist unexpected inputs such as imaging artifacts, unique patient positioning or size, or surgical changes, by using task-appropriate guardrails to ensure that explanations remain medically plausible (e.g., anatomic constraints during model training) [25,26].
Design XAI for adaptive personalization to end users.	Individual end users have their own unique preferences and cognitive styles. Personalization of XAI is critical to optimizing human-AI collaboration. Adaptive personalization will be key in preventing these XAI tools from remaining static and suboptimally tailored for individuals.	- Leverage real-time user preference data collected during app usage through reinforcement learning techniques (e.g., RLHF, DPO) to customize the style of generated content (e.g., clinical notes and radiology reports) [18]. - Adjust outputs depending on subspecialty: selectively focus on features pertinent to subspecialty-dependent decision points (e.g., surgical vs nonsurgical imaging findings). - Adjust outputs depending on experience level (resident, fellow, attending) [16], for example providing more nuanced explanations for attendings compared to interns. - Evaluate the necessity of adaptation; reserve customizable models for complex tasks with diverse user groups while using reliable static models for straightforward binary tasks.
Match XAI choices to clinical tasks and outputs.	Different clinical tasks require different XAI outputs. Alignment of explanation formats with clinical expectations and use cases enhances the clinical applicability of these methods.	- Classification: For binary or multilabel model predictions, include localization (e.g., saliency maps, bounding boxes) or quantitative explanations if the abnormality has pre-defined thresholds (e.g., definition of an ascending aortic aneurysm based on a diameter > 5 cm). - Regression: Integrate multiple data points (e.g., patient age, presence or absence of diabetes, blood pressure) to predict a continuous variable (e.g., 5-year stroke risk) [3,5]; include semantic (e.g., diabetes status) and counterfactual (e.g., adjustment of patient age or blood pressure to demonstrate changes in predicted stroke risk) explanations when applicable.

RLHF= reinforcement learning with human feedback, DPO= Direct preference optimization.

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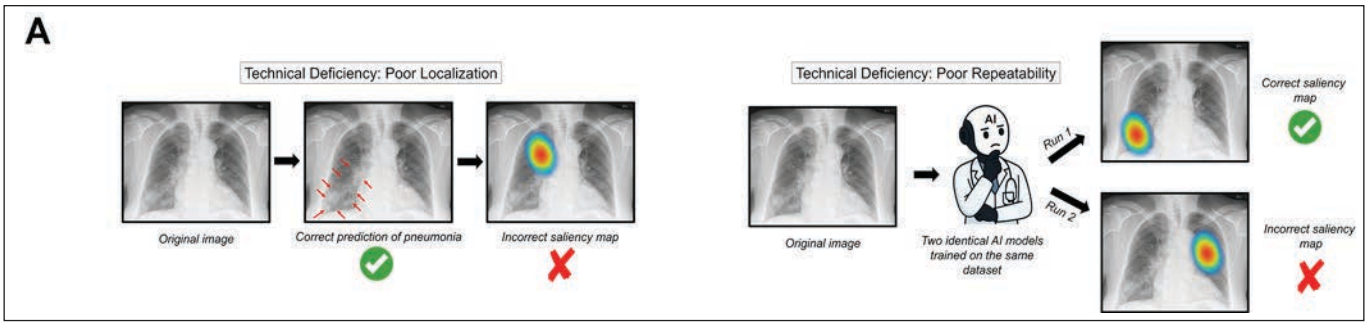


Figure 1. Explainable artificial intelligence (XAI) methods in medical imaging should be technically robust, adapt to end users, and provide explanations that match clinical task. (A) Diagram shows two examples of XAI that lacks technical robustness. XAI with poor localization correctly classifies presence of pneumonia but incorrectly highlights hilar vasculature instead of right lower lobe consolidation. (B) Diagram shows example of three end users (radiologist, surgeon, patient) receiving results of intracranial hemorrhage tailored to their specific needs or expertise. Preference data collected via feedback mechanisms (e.g., surveys, ranked model outputs) can be used to adapt XAI model outputs to user preferences using existing reinforcement learning architectures such as reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO). (C) Diagram shows examples of three clinical use cases and their corresponding localization, quantitative, and semantic-based XAI methods. Use cases for which focal imaging features (e.g., consolidation) explain artificial intelligence (AI) output (e.g., pneumonia) may benefit most from localization. Quantitative XAI, in comparison, may be best suited for use cases for which medical literature has established specific thresholds (e.g., presence of ascending aorta diameter that exceeds 5 cm to establish diagnosis of aneurysm). Semantic XAI may help provide clinically relevant contextual meaning (e.g., closed vs open growth plate) to model predictions that exist along spectrum (e.g., bone age). Imagen (version 4.0, Google DeepMind) was used to generate initial images for AI robot of Figure 1A and 1B, "Radiologist", "Surgeon", and "Patient", of Figure 1B, and aorta/heart and hand of Figure 1C. These images were then further manually modified for stylistic preference.

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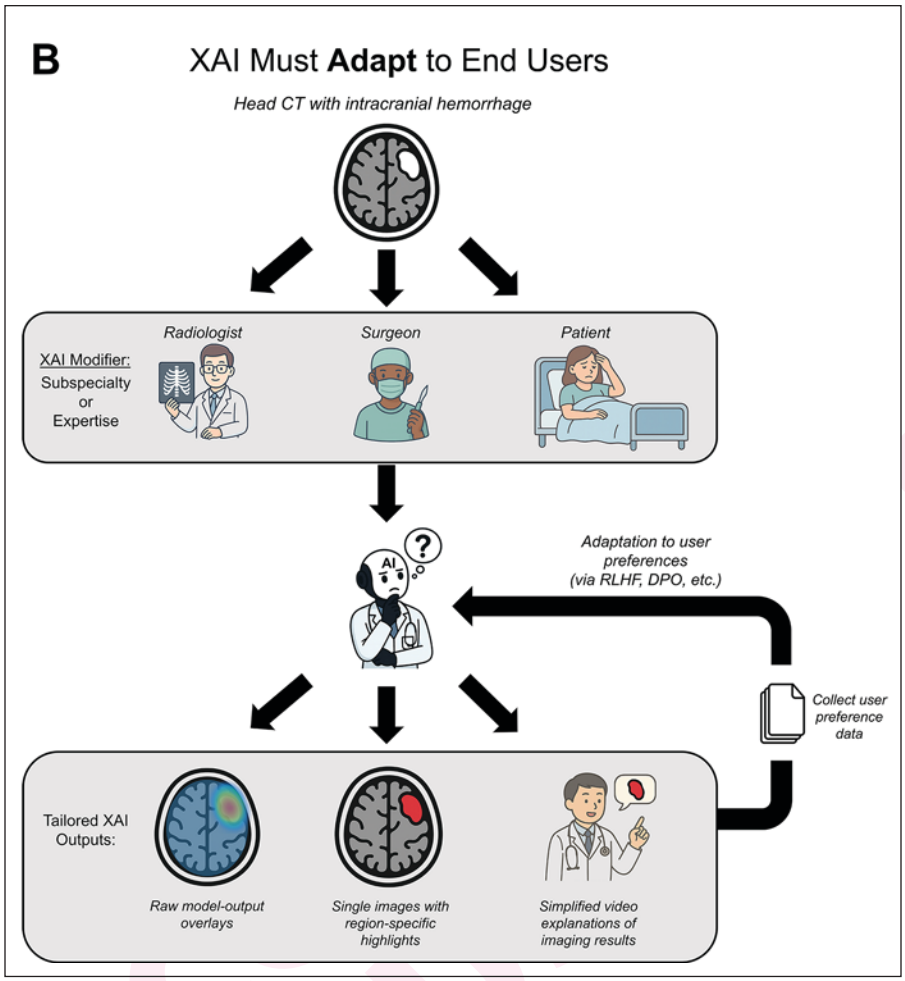


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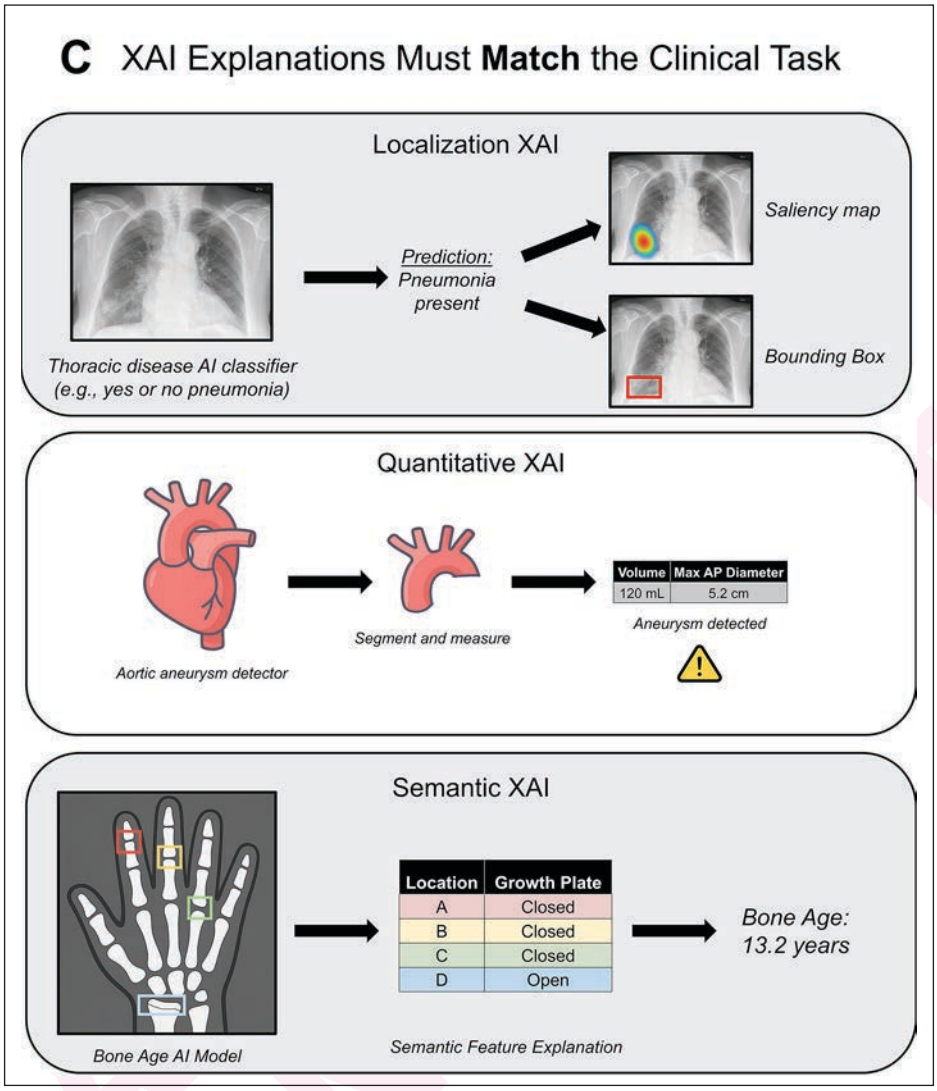


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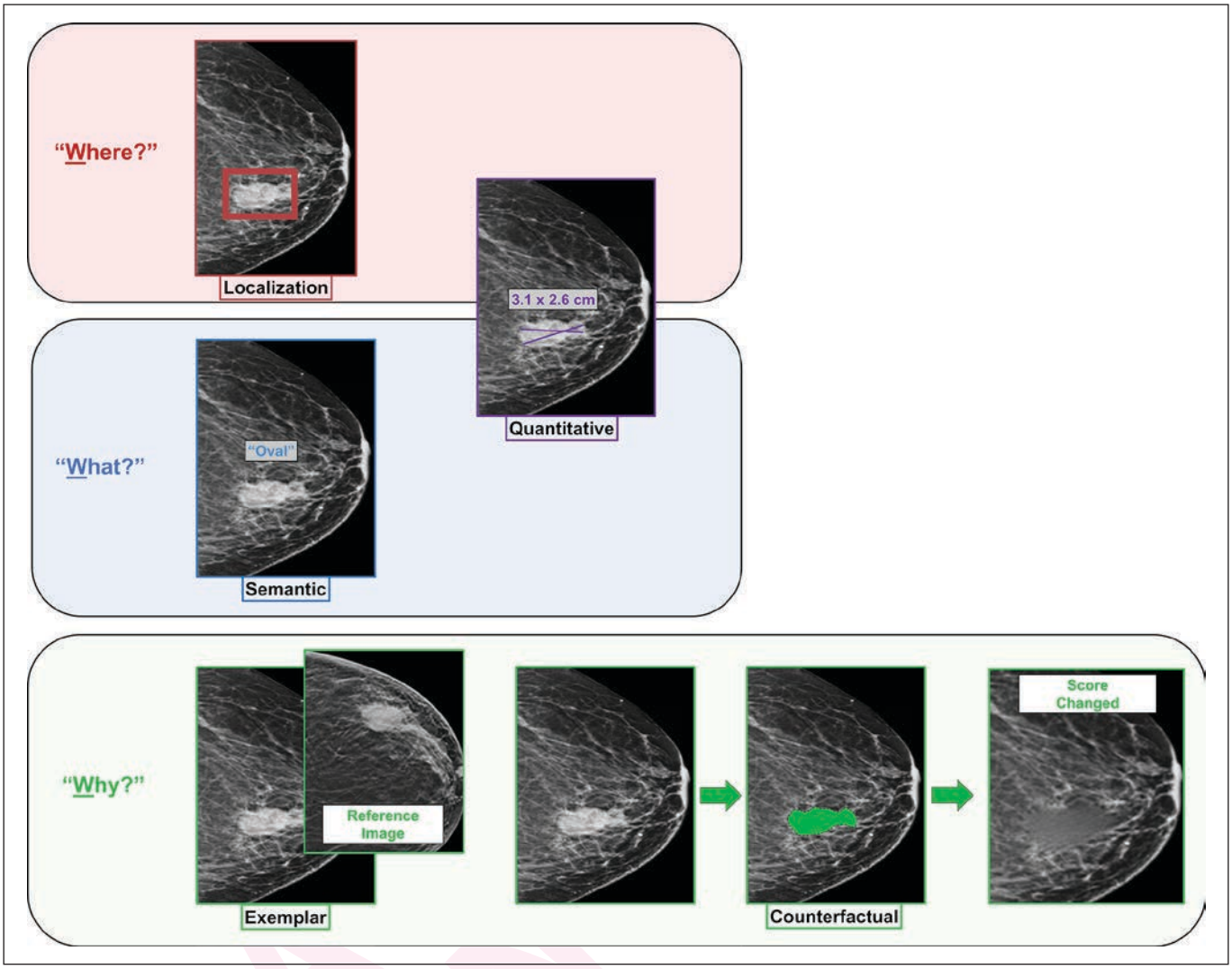


Figure 2. Three fundamental questions regarding the manner in which different types of explainable artificial intelligence (XAI) seek to answer questions about model outputs: where is abnormality located, what is abnormal, and why is it abnormal. Localization techniques such as bounding boxes (red rectangle), attempt to answer "where" question by identifying small locations in image that strongly contributed to model's output. Other XAI methods, like semantic XAI (blue rectangle), attempt to answer "what" question by describing characteristics of abnormality. More recently, exemplar XAI methods (green rectangle) have targeted "why" question by showing example reference images in training dataset that are similar to current image, allowing users to reach their own conclusions regarding rationale for model prediction. This type of XAI may be more useful for complex model outputs with many important predictive features. Similarly, counterfactual XAI (green rectangle) answers "why" question by allowing user to modify image (e.g., removal of breast lesion) in order to compare and contrast changes in model output. Some XAI types, such as quantitative XAI (purple rectangle), may answer more than one question. Use of these XAI types varies among FDA-cleared AI tools in medicine. (Adapted from McNamara SL, Yi PH, Lotter W. The clinician-AI interface: intended use and explainability in FDA-cleared AI devices for medical image interpretation, NPJ Digit Med 2024; 7:80, © 2024, with permission of the authors under the terms of Creative Commons Attribution 4.0 International Public License (https://creativecommons.org/licenses/by/4.0/legalcode), including the disclaimer of warranties in Section 5)

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Explainable Artificial Intelligence (AI) for Medical Imaging: A Framework for Bridging the AI Trust Gap

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Paul H Yi: Conception and design of the manuscript and substantively revised it.

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